

V S K

ELECTRO-MECH SOLUTIONS

SPECIAL PURPOSE MACHINE BUILDERS

CNC Reconditioning · Industrial Automation · Precision Components
Hydraulic & Pneumatic Design · Mechanical Design · Industrial Spares



CORPORATE PROFILE

Engineering Excellence Since February 2011

300+ Machines Designed, Manufactured & Delivered Pan-India

No. 834, 8th Cross, 3rd Phase, Ganapathinagar, Peenya Industrial Area, Bangalore — 560058

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— ABOUT US

VSK Electro-Mech Solutions is a Bangalore-based engineering enterprise established in February 2011. We specialize in the design, manufacturing, and supply of Special Purpose Machines (SPMs), CNC machine reconditioning and retrofitting, hydraulic and pneumatic system design and manufacturing, mechanical machine design, precision component manufacturing, industrial automation, and supply of electrical and mechanical industrial spares.

As a registered Small Scale Industry (SSI) unit operating from a well-equipped 1,700 sq. ft. facility in Peenya Industrial Area — India's largest industrial estate — we have grown into a trusted partner for leading manufacturers across the country. With over 300 custom-engineered machines successfully operational at client facilities nationwide, our track record speaks for itself.

Our leadership team comprises qualified engineers with deep experience gained at senior managerial levels in large, reputed manufacturing organizations. This expertise spans mechanical design, CNC technology, PLC/HMI programming, hydraulics, pneumatics, and machine electrification — enabling us to deliver complete, turnkey solutions for the most demanding production challenges.

300+ MACHINES DELIVERED	2011 ESTABLISHED	8 BUSINESS DIVISIONS	40+ SPM VARIANTS
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Company Information

Company Name	M/s VSK Electro-Mech Solutions
Date of Incorporation	February 2011
Constitution	Proprietorship / Partnership / Private Ltd.
Industry Type	Small Scale Industry (SSI) — Registered Unit
Registered Address	No. 834, 8th Cross, 3rd Phase, Ganapathinagar, Peenya Industrial Area, Bangalore — 560058
Factory Area	1,700 Sq. Ft.
Connected Power	40 HP
PAN	ABHPPH4480R
TIN	29970601703
Contact — Mr. Shekara H S	+91 98803 36714 / +91 73531 00095
Contact — Mr. Mohan Kumar	+91 94488 73708
Email	vsk.electromech@gmail.com

Team Strength

Our dedicated, skilled workforce is structured for rapid project execution:

- Managers / Project Leads — 2
- Electricians — 4
- Fitters / Assembly Technicians — 2
- PLC & CNC Programmers — 1
- Additional contract specialists engaged for large-scale and multi-site projects

— WHAT WE DO — CORE BUSINESS DIVISIONS

VSK Electro-Mech Solutions operates across seven interconnected business divisions, delivering end-to-end engineering solutions under one roof.

DIVISION 1

Special Purpose Machine (SPM) Design, Manufacturing & Supply

Our flagship capability — engineering custom SPMs from concept to commissioning. We handle the complete lifecycle: requirement analysis, mechanical design (SolidWorks), hydraulic and pneumatic system design, electrical panel design and manufacturing, PLC/HMI programming, fabrication, assembly, testing, and on-site installation and training.

Our SPM portfolio includes CNC turning and boring machines, notching machines (2-axis and 4-axis servo), cross-cutting machines, ring cutting machines, conveyor systems with PLC interfacing, vision inspection systems, gantry loading/unloading systems, leakage testing machines, deburring machines, drilling and chamfering machines, greasing and painting machines, pressing machines, honing machine control panels, and milling machines.

DIVISION 2

CNC Machine Reconditioning & Retrofitting

We restore and upgrade high-value CNC machinery from leading global manufacturers, returning them to peak performance. Each reconditioning project includes complete mechanical overhaul, electrical system replacement or upgrade, CNC control migration, rigorous testing, and re-commissioning. Our expertise covers:

- Cylindrical grinding machines — Studer, Kellenberg, Toyoda, Mitsubishi, Voumard, Jung Berlin, Koyo
- Machining centres — HMC (60-tool magazine), VMC (HMT make)
- CNC turning centres — HMT, Sands, Hardinge, Hyundai
- Jig grinding machines — Hauser 5-axis, retrofitted with PLC, HMI & 3-axis servo motors
- Vertical honing machines — 4-axis PLC/HMI retrofit
- CNC control platforms — Fanuc Oi-TF, Fanuc Oi-TF Plus

DIVISION 3

Precision Component Manufacturing

Leveraging our in-house CNC machining, grinding, and assembly capabilities, we manufacture precision-engineered components, jigs, fixtures, and tooling to exacting tolerances. Our machines routinely achieve dimensional accuracy of ± 0.1 mm down to 0.02 mm, and our wheel balancing systems hold 20-micron face-out precision.

Our precision manufacturing serves industries including:

- Automotive components and sub-assemblies
- Hydraulic system components (cylinders, spools, valves)
- Sealing system parts — metal seals, PTFE tubes, backup rings, bushings

- Gear manufacturing support — motor spigots, motor flanges, shafts
- Custom jigs, fixtures, and inspection gauges for production lines
- General engineering — shafts, spindles, mandrels, adapters, and wear parts

DIVISION 4**Hydraulic & Pneumatic System Design and Manufacturing**

We offer complete hydraulic and pneumatic system design and manufacturing services — from concept and circuit design through to fabrication, assembly, testing, and integration with SPM machines and production equipment:

- Hydraulic system design, analysis, and manufacturing for SPM machines and industrial equipment
- Pneumatic system design, circuit layout, and manufacturing
- Hydraulic power pack design and assembly
- Hydraulic pressing systems — custom-built for bearing pressing, seal pressing, and component assembly
- Pneumatic actuator and cylinder integration for automated machine operations
- Hydraulic and pneumatic system troubleshooting, problem analysis, and repair
- Air pipeline installation and maintenance works

DIVISION 5**Mechanical Machine Design and Manufacturing**

Our mechanical design capability is central to everything we build. Using SolidWorks CAD platform, we take each machine from initial concept through detailed 3D modelling, structural analysis, and manufacturing drawings — ensuring every design is optimised for precision, durability, and ease of maintenance:

- Complete SPM machine design using SolidWorks — 3D modelling, assemblies, and production drawings
- Fabricated and cast bed structures designed for rigidity and vibration damping
- Machine alignment correction and precision levelling
- Machine installation and commissioning at customer sites
- Complete machine reconditioning — slideway regrounding, spindle rebuilds, and structural restoration
- Installation and maintenance of multi-spindle machines and SPMs

DIVISION 6**Industrial Automation & Control Systems**

We design and implement complete automation solutions integrating PLC, HMI, VFDs, and vision inspection systems to transform manual operations into fully automated production lines:

- PLC/HMI controlled automation — Delta, Mitsubishi, Siemens platforms
- Servo-controlled pick-and-place systems and gantry loading/unloading
- Auto-loading and stacker systems with VFD-controlled motors
- Vision inspection integration for quality verification
- Conveyor automation with PLC interfacing for grinding and machining lines

- Touch screen / HMI programming for operator-friendly machine control

DIVISION 7

Electrical Installations & Panel Manufacturing

Our electrical division provides end-to-end machine electrification, control panel manufacturing, and CNC cable harnessing services:

- Machine control panel design, manufacturing, and wiring per customer specifications
- LT panel manufacturing and supply
- CNC machine cable harnessing — complete cable harness design, routing, and termination for all types of CNC machines
- Electrical harness design and manufacturing for SPM machines and industrial equipment
- Machine control circuit design and problem rectification
- Reconditioning of all types of machine control panels

DIVISION 8

Electrical & Mechanical Industrial Spares

We supply a comprehensive range of electrical and mechanical industrial spares to keep your production lines running with minimal downtime:

- Electrical spares — contactors, relays, overload relays, MCBs, switches, sensors, proximity switches, limit switches, encoders, and cable accessories
- Mechanical spares — bearings, couplings, belts, pulleys, gears, seals, O-rings, bushings, spindle components, and wear parts
- CNC machine spares — spindle bearings, tool holders, chuck jaws, way wipers, bellows, and coolant system components
- PLC/HMI/VFD spares — Delta, Mitsubishi, Siemens modules, power supplies, and communication cards
- Pneumatic and hydraulic spares — cylinders, solenoid valves, FRL units, pressure gauges, hoses, and fittings
- Prompt sourcing and supply from OEM and quality aftermarket channels across India

– TECHNICAL CAPABILITIES

Our multi-disciplinary engineering team brings together expertise across four core engineering domains, enabling us to deliver integrated solutions that no single-discipline shop can match.

Mechanical Engineering

- SPM machine design and manufacturing — concept to delivery
- Machine installation, alignment correction, and precision levelling
- Complete machine reconditioning and retrofitting of all types of CNC & conventional machines
- Installation & maintenance of multi-spindle machines and SPMs
- Mechanical design using SolidWorks CAD platform

Hydraulic & Pneumatic Engineering

- Hydraulic system design, analysis, troubleshooting, and repair
- Pneumatic system design, analysis, and problem solving
- Air pipeline installation and maintenance
- Custom hydraulic and pneumatic pressing systems design and build

CNC & Electronic Control Systems

- PLC/HMI programming — Delta, Mitsubishi, Siemens
- CNC control systems — Fanuc
- Servo-controlled automation — pick-and-place, gantry systems, auto-indexing mechanisms
- Vision inspection system integration
- VFD-controlled spindle and motor drive systems
- Touch screen / HMI programming and integration

Electrical Engineering

- Machine control panel design, fabrication, wiring, and reconditioning
- CNC machine cable harnessing — design, routing, and termination
- PLC machine and CNC machine control panel wiring
- LT panel design, installation, servicing, and AMC
- Machine control circuit design and problem rectification

— SPM MACHINES — COMPLETE PORTFOLIO

Over 40 distinct SPM machine variants designed, manufactured, and delivered to customers across India. Each machine is custom-engineered to solve a specific production challenge — delivering measurable improvements in quality, throughput, and operational efficiency.



Servo/PLC Controlled CNC Lathe · CNC Machine with Full Enclosure

A. Turning, Boring & Facing Machines

- 01 Vertical Turning CNC Lathe — 900 mm length PTFE rod processing
- 02 Horizontal PLC & Servo-Controlled 2-Axis Lathe
- 03 Vertical Rod Boring Machine — 350 mm dia, 6-Jaw Chuck, Fanuc CNC, 2-axis, 5 HP spindle
- 04 Motor Spigot CNC Turning Machine
- 05 Motor Flange Facing Machine
- 06 Vertical Type Seal Facing Machine
- 07 Metal Facing Machine — casting bed, pneumatic axis, MT4 tailstock, 5 HP, 12 sec cycle

B. Notching & Cutting Machines

- 08 2-Axis Servo Notching Machine
- 09 4-Axis Servo Notching Machine
- 10 4-Axis Servo Notching Machine — simultaneous top and bottom notch
- 11 4-Axis Servo Notching Machine with Auto Loading & Stacker System
- 12 2-Axis Bottom Notching Machine with O-Ring Assembly & Vision Inspection
- 13 Razer Cross Cutting Machine for Backup Rings
- 14 Seal Cross Cutting Machine with Auto Loading System
- 15 Ring Cutting Machine — 50 mm to 200 mm with Auto Loading
- 16 Bush Cross Cutting Machine with Auto Loading
- 17 Rubber Bush Cutting Machine — 30 mm to 50 mm range
- 18 Z-Cut Tercite Punching Machine — 0.02 mm tolerance
- 19 Spiral Mandrel Cutting Machine



Metal Facing Machine — 12-Second Cycle Time for Seal Finishing

C. Assembly, Inspection, Testing & Finishing

- 20 Air Leakage Testing Machine with Festo Servo Unit
- 21 Aluminium Component Deburring Machine
- 22 Thread Deburring Machine
- 23 Drilling & Chamfering Machine with Auto Pick-Up and Unloading
- 24 Greasing Machine for Seals
- 25 Seal Scrubbing and Adhesive Spraying Machine
- 26 Seal Scrubbing and Polishing Machine with Auto Indexing
- 27 Internal Seal Scrubbing and Spraying Machine
- 28 Epoxy Painting Machine for Seals
- 29 Manual Painting, Adhesive Dipping & Drying Machine
- 30 Seal Pressing Machine
- 31 Spinning Machine for Metal Seals
- 32 Paint Agitating Machine

D. Conveyor & Material Handling Systems



PLC-Interfaced Auto-Loading Conveyor for Centreless Grinding



PLC-Interfaced Conveyor Systems for Centreless Grinding Machines

- 33 Conveyor for Centreless Grinding Machine — PLC-interfaced auto loading and feeding
- 34 Food Grade Conveyor for Cashew Industry
- 35 Wheel Balancing Unit — 1000 mm dia grinding wheel, 20-micron face-out accuracy

E. Pressing & Hydraulic Machines

- 36 Hydraulic Pressing Machine for Bearing Pressing to Rotor Shaft
- 37 Hand Pressing Machine for Bearing Pressing

F. Milling & Boring SPMs

- 38 2-Meter Conveyor Rod Milling Machine — up to 60 mm diameter
- 39 2-Meter, 200 mm Dia Conveyor Pipe Boring Machine — dual spindle

G. Control Panels & Systems

- 40 Gehring Maske Honing Machine Control Panel — complete design and manufacturing
- 41 Custom Hydraulic & Pneumatic System Design and Manufacturing

— CNC RECONDITIONING & RETROFIT PORTFOLIO

Proven expertise in restoring and upgrading high-value CNC machinery from the world's leading machine tool manufacturers. Each project includes complete mechanical overhaul, electrical system replacement or upgrade, CNC control migration, precision testing, and re-commissioning with performance validation.

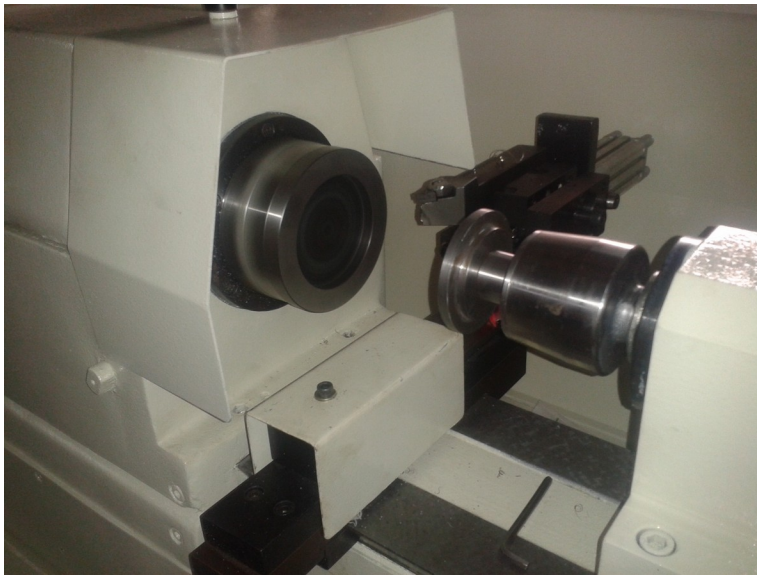


CNC Vertical Boring Machine with Fanuc Panel · CNC Rod Boring Machine

Grinding Machines Reconditioned

- 01 Studer OD CNC Grinding Machine — Fanuc Oi-TF CNC Control
- 02 Mitsubishi Grinding Machine — Fanuc Oi-TF CNC Control
- 03 Toyoda Grinding Machine — Fanuc Oi-TF Plus System
- 04 Koyo Internal Grinding Machine — Fanuc Oi-TF Plus CNC Control
- 05 Jung Berlin CNC Grinding Machine — Fanuc Oi-TF Plus CNC Control
- 06 Kellenberg CNC Grinding Machine — Fanuc Oi-TF Plus System
- 07 Voumard CNC Internal Grinding Machine — Fanuc Oi-TF Plus System

Machining Centres Reconditioned



Precision Spindle and Chuck Assembly — Reconditioning Quality Standard

- 08 HMC Machine — Fanuc Control with 60-Tool Magazine
- 09 HMT Make VMC Machine — Full Reconditioning
- 10 HMT Make Turning Centre — Complete Overhaul

Turning Machines Reconditioned

- 11 Sands Flat Bed Turning Lathe — Fanuc CNC Control Retrofit
- 12 Hardinge Machine — Mechanical Retrofitting
- 13 Hyundai CNC Turning Machine — Reconditioning

Specialty Machine Retrofits

- 14 Hauser 5-Axis Jig Grinding Machine — Retrofitted with PLC, HMI & 3-Axis Servo Motors
- 15 4-Axis Vertical Honing Machine — Retrofitted with PLC & HMI Unit

— FEATURED PROJECT CASE STUDIES

Representative projects showcasing our engineering capabilities across SPM design, CNC integration, and precision manufacturing.

Project 1 — Table Top CNC Lathe (Servo/PLC Controlled)

Client: Trelleborg Sealing Systems, Hosur Road, Bangalore

Application: Pre-turning of 850 mm long PTFE rubber rods (dia 12 mm to 120 mm) to a tolerance of ± 0.1 mm.

Technical Specs: Fabricated bed, servo-controlled axes, PLC logic, touch screen HMI programming.



Table Top CNC Lathe — Servo and PLC Controlled with Touch Screen HMI

Project 2 — CNC Rod Boring Machine (Fanuc CNC)

Client: Trelleborg Sealing Systems, Hosur Road, Bangalore

Application: Pre-boring of 350 mm long PTFE rubber tubes (dia 350 mm to 1000 mm) to ± 0.1 mm tolerance.

Technical Specs: Casting bed, Fanuc 2-axis CNC controller, VFD-controlled spindle, 250 mm dia chuck, 5 HP spindle power.

Project 3 — PLC-Interfaced Conveyor Systems

Client: TMI Rollon Engineering Pvt. Ltd.

Application: Automated loading and feeding of spool components to centreless grinding, Lidcoping grinding, and HMT grinding machines. Eliminated manual loading bottlenecks and significantly improved grinding throughput.

Technical Specs: Fabricated structure, geared and VFD-controlled motor drive, PLC interfacing.

Project 4 — Metal Seal Facing Machine

Client: Trelleborg Sealing Systems, Hosur Road, Bangalore

Application: Final finishing operation on metal oil seals after pressing. Achieves a cycle time of just 12 seconds per component.

Technical Specs: Casting bed, pneumatic sliding axis, MT4 tailstock (180 mm stroke), 5 HP spindle motor.

Project 5 — Wheel Balancing Unit (20-Micron Precision)

Client: Trelleborg Sealing Systems, Hosur Road, Bangalore

Application: External alignment of 1000 mm diameter grinding wheels with 20-micron face-out accuracy.

Technical Specs: Fabricated structure with precision-ground wheel mounting surface for the wheel spindle.

Project 6 — Dual-Spindle Pipe Boring Machine (2-Meter)

Client: Sapthagiri Enterprises, Salem

Application: Boring of 200 mm diameter pipes up to 2 meters in length using dual spindle configuration for high-throughput production.

— KEY CUSTOMERS & INDUSTRY PARTNERS

We are proud to serve some of India's most demanding manufacturers, delivering mission-critical machines and services that are integral to their production operations.

CUSTOMER	SCOPE OF WORK
Trelleborg Sealing Systems India Pvt. Ltd.	30+ machine variants — metal seals, SPMs, jigs & fixtures, CNC boring, notching, cutting, deburring, painting, assembly, and inspection machines
Yuken India Limited	Hydraulic components — machine servicing, retrofitting, and electrical installations
Transtech Gears India Pvt. Ltd.	Motor spigot CNC turning, motor flange facing, hydraulic & hand pressing machines for gear manufacturing
TMI Rollon Engineering Pvt. Ltd.	PLC-interfaced conveyor systems for centreless grinding machine auto loading and feeding
Sapthagiri Enterprises, Salem	2-meter rod milling machine (up to 60 mm dia), 2-meter dual-spindle pipe boring machine (200 mm dia)
Shiva Industry (Cashew)	Food-grade conveyor systems for cashew processing industry

— WHY CHOOSE VSK ELECTRO-MECH SOLUTIONS

WHAT SETS US APART

- **Proven Track Record** — 300+ machines designed, manufactured, and successfully running at customer sites across India.
- **End-to-End Capability** — From concept design in SolidWorks to PLC programming, fabrication, assembly, testing, and on-site commissioning.
- **Multi-Disciplinary Expertise** — Mechanical, hydraulic, pneumatic, electrical, CNC, PLC/HMI, and VFD systems under one roof.
- **CNC & PLC/HMI Expertise** — Fanuc CNC systems and Delta, Mitsubishi, Siemens PLC/HMI platforms.
- **Precision Engineering** — Machines achieving tolerances of ± 0.1 mm to 0.02 mm, and 20-micron face-out accuracy.
- **Precision Component Manufacturing** — In-house CNC machining of jigs, fixtures, tooling, and precision parts for automotive, hydraulics, sealing, and gear industries.
- **Rapid Turnaround** — Lean team with experienced leadership enables fast project delivery from enquiry to commissioning.
- **Customer-Centric Service** — Dedicated service team for after-sales support, AMC, and ongoing maintenance.
- **Cost-Effective Solutions** — SSI-registered unit offering competitive pricing without compromising on engineering quality.
- **Industry Diversity** — Serving hydraulics, sealing systems, gears, automotive, food processing, and general engineering sectors.

LET'S ENGINEER YOUR NEXT SOLUTION

Whether you need a custom SPM, CNC reconditioning, hydraulic/pneumatic systems, precision components, automation, or industrial spares — we're ready.

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